MA 324

STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS

IIT GUWAHATI

END-SEMESTER EXAMINATION

14:00–17:00 IST

MAY 8, 2026

PART-A

Name of the Student: B. Hemanth Naik

Signature of the Student: Hemanth

Roll No.: 230123018

Signature of the Invigilator:

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	0	9	14

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3,$ be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

The objective function $\cancel{\phi}$ $Q(\theta_1,\theta_2) = \sum_{i=1}^{n} \varepsilon_i^2$

$$\phi(\theta_1,\theta_2) = (Y_1-\theta_1-\theta_2)^2 + (Y_2-\theta_1+2\theta_2)^2 + (Y_3-2\theta_1+\theta_2)^2$$

To get the normal equation differentiate w.r.t θ_1 and θ_2 and equate to 0.

$$\frac{\partial}{\partial\theta_1}\phi(\theta_1,\theta_2) = 2(Y_1-\theta_1-\theta_2)(-1) + 2(Y_2-\theta_1+2\theta_2)(-1) + 2(Y_3-2\theta_1+\theta_2)(-2) = 0$$

$$Y_1-\theta_1-\theta_2 + Y_2-\theta_1+2\theta_2 + 2Y_3-4\theta_1+2\theta_2 = 0$$

$$Y_1+Y_2+2Y_3-6\theta_1+3\theta_2 = 0$$

$$\boxed{\hat{\theta}_1 = \frac{Y_1+Y_2+2Y_3+3\hat{\theta}_2}{6}}$$

$$\frac{\partial}{\partial\theta_2}\phi(\theta_1,\theta_2) = 2(Y_1-\theta_1-\theta_2)(-1) + 2(Y_2-\theta_1+2\theta_2)(2) + 2(Y_3-2\theta_1+\theta_2) = 0$$

$$-Y_1+\theta_1+\theta_2 + 2Y_2-2\theta_1+4\theta_2 + Y_3-2\theta_1+\theta_2 = 0$$

$$-Y_1+2Y_2+Y_3-3\theta_1+6\theta_2 = 0$$

$$\hat{\theta}_2 = \frac{Y_1-2Y_2-Y_3+3\theta_1}{6}$$

$$6\theta_2 = Y_1-2Y_2-Y_3+\cancel{3}\left(\frac{Y_1+Y_2+2Y_3+3\theta_2}{\cancel{6}2}\right)$$

$$12\theta_2 = 2Y_1 - 4Y_2 - \cancel{2Y_3} + Y_1 + Y_2 + \cancel{2Y_3} + 3\theta_2$$

$$9\theta_2 = 3Y_1 - 3Y_2$$

$$\boxed{\hat{\theta}_2 = \frac{Y_1 - Y_2}{3}}$$

$$6\hat{\theta}_1 = Y_1 + Y_2 + 2Y_3 + 3\hat{\theta}_2$$

$$= Y_1 + Y_2 + 2Y_3 + \cancel{3}\left[\frac{Y_1 - Y_2}{\cancel{3}}\right]$$

$$6\hat{\theta}_1 = 2Y_1 + 2Y_3$$

$$\boxed{\hat{\theta}_1 = \frac{Y_1 + Y_3}{3}}$$

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

sample Covariance $S = \begin{bmatrix} var(x_1) & Cov(x_1,x_2) & Cov(x_1,x_3) \\ Cov(x_2,x_1) & var(x_2) & Cov(x_2,x_3) \\ Cov(x_3,x_1) & Cov(x_3,x_2) & var(x_3) \end{bmatrix}$

density function for U_1 $f(x_1) = \frac{1}{b-a} = \frac{1}{1} = 1$

lly U_2 $f(x_2) = 1$

$E[x_1] = \int_0^1 x_1 dx_1 = \frac{x_1^2}{2} = \frac{1}{2}$

$E[x_2] = \int_0^1 x_2 dx_2 = \frac{1}{2}$

$E[x_1^2] = \int_0^1 x_1^2 dx_1 = \frac{1}{3}$

$E[x_2^2] = \frac{1}{3}$ and $E[x_3] = E[x_1] + E[x_2] = 1$

$var(x_1) = E[x_1^2] - (E[x_1])^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

$var(x_2) = \frac{1}{12}$

$var(x_3) = var(x_1) + var(x_2) + 2cov(x_1,x_2) = \frac{1}{12} + \frac{1}{12} + 2Cov(x_1,x_2)$

$Cov(X,Y) = E[(X - E(X))(Y - E(Y))]$

$Cov(x_1,x_2) = E\left[\left(x_1 - \frac{1}{2}\right)\left(x_2 - \frac{1}{2}\right)\right] = E\left[x_1x_2 - \frac{x_1}{2} - \frac{x_2}{2} + \frac{1}{4}\right]$

$f_{x_1x_2}(x_1,x_2) = f_{x_1}(x_1) f_{x_2}(x_2) = 1 \cdot 1 = 1$

$E[x_1 x_2 = x] = \int_0^1 x\, dx = \left(\frac{x^2}{2}\right)_0^1 = \frac{1}{2}$

$Cov(x_1,x_2) = \frac{1}{2} - \frac{1}{2}\times\frac{1}{2} - \frac{1}{2}\times\frac{1}{2} + \frac{1}{4} = \frac{1}{4}$ ✗

$Cov(x_1,x_3) = var(x_1) + var(x_2) + 2Cov(x_1,x_3) = \frac{1}{12} + \frac{2}{3} + 2Cov(x_1,x_3)$

$= \frac{1}{12} + \frac{2}{3} + 2$ ✗

$var(x_3) = \frac{1}{12} + \frac{1}{12} + \cancel{2}\cdot\frac{1}{\cancel{4}2} = \frac{2}{3}$

$Cov(x_1, x_3) = E[(x_1 - \frac{1}{2})(x_3 - 1)] = E[x_1 x_3 - x_1 - \frac{x_3}{2} + 1]$

$= \frac{1}{2} - \frac{1}{2} - \frac{1}{2} + 1$

$= \frac{1}{2}$

$$S = \begin{bmatrix} \frac{1}{12} & \frac{1}{4} & \frac{7}{4} \\ \frac{1}{4} & \frac{1}{12} & \frac{7}{4}_{2} \\ \frac{7}{4} & \frac{7}{4} & \frac{2}{3} \end{bmatrix}$$

$r_{ij} = \frac{S_{ij}}{\sqrt{S_{ii}}\sqrt{S_{jj}}}$

$$r = \begin{bmatrix} 1 & 3 & \frac{21}{2\sqrt{2}} \\ 3 & 1 & \frac{21}{2\sqrt{2}} \\ \frac{21}{2\sqrt{2}} & \frac{21}{2\sqrt{2}} & 1 \end{bmatrix}$$

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b) $|r - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 3 & \frac{21}{2\sqrt{2}} \\ 3 & 1-\lambda & \frac{21}{2\sqrt{2}} \\ \frac{21}{2\sqrt{2}} & \frac{21}{2\sqrt{2}} & 1-\lambda \end{vmatrix} = 0 \Rightarrow \cancel{1-3\left[\frac{21\times 21}{4\times 2}\right] + \frac{21}{2\sqrt{2}} \times \frac{3\times 21}{1\cdot 2\sqrt{2}}}$$

$$(1-\lambda)\left[(1-\lambda)^2 - \frac{21\times 21}{4\times 2}\right] - 3\left(3(1-\lambda) - \frac{21\times 21}{4\times 2}\right) + \frac{21}{2\sqrt{2}}\left(\frac{3\times 21}{2\sqrt{2}} - \frac{21(1-\lambda)}{2\sqrt{2}}\right) = 0$$

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3. Let the matrix of distance be

$$\begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:
   i. Complete linkage hierarchical procedure.
   ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

a) i) In complete linkage hierarchial procedure.

$$d_{(UV)W} = \max\{d_{U,W}, d_{V,W}\}$$

The minimum distance is between items 1 and 2 i.e 1

cluster 1 = {1, 2} , height = 1 = distance.

	1, 2	3	4
1, 2	0		
3	11	0	
4	5	3.5	0

$$d_{(1,2)(3)} = \max\{d_{1,3}, d_{2,3}\} = \max\{11, 2\} = 11$$

$$d_{(1,2)(4)} = \max\{d_{1,4}, d_{2,4}\} = \max\{5, 3\} = 5$$

Now. the minimum distance is between items 4 & 3 i.e 3.5

cluster 2 = {3, 4} height = 3.5

	1, 2	3, 4
1, 2	0	
3, 4	11	0

$d_{(1,2)(3,4)} = \max\{d_{(1,2),3}, d_{(1,2)(4)}\} = \max\{11, 5\} = 11$

Now the minimum distance is betwen items (3,4) & (1,2)

4 clusters

	1234
1234	0

cluster 3 = {1,2,3,4} height 11

ii) In Average linkage hierarchial prodocure

$$d_{A,B} = \frac{1}{|A||B|} \sum_{i\in A} \sum_{j\in B} d_{i,j}$$

|A| = no. of elements or items in cluster A

|B| = no. of items in cluster B

The minimum distance is between 1 & 2 i.e 1

cluster 1 = {1,2} height = 1

	1,2	3	4
1,2	0		
3	6.5	0	
4	4	3.5	0

$$d_{(1,2)(3)} = \frac{1}{2\times 1} \sum_{i\in(1,2)} d_{i,3} = \frac{d_{1,3}+d_{2,3}}{2} = \frac{11+2}{2} = 6.5$$

$$d_{(1,2)(4)} = \frac{1}{2\times 1} \sum_{i\in(1,2)} d_{i,4} = \frac{d_{1,4}+d_{2,4}}{2} = \frac{5+3}{2} = 4$$

Now, the minimum distance is between 4 & 3 i.e 3.5

cluster 2 = {3,4} height = 3.5

	1,2	3,4
1,2	0	
3,4	5.25	0

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Additional Pages for Answers

3) Continuation.

$$d_{(1,2)(3,4)} = \frac{1}{2\times 2} \sum_{i\in(1,2)} d_{i,3}+d_{i,4} = \frac{d_{1,3}+d_{1,4}+d_{2,3}+d_{2,4}}{4}$$

$$= \frac{11+5+2+3}{4}$$

$$= 5.25$$

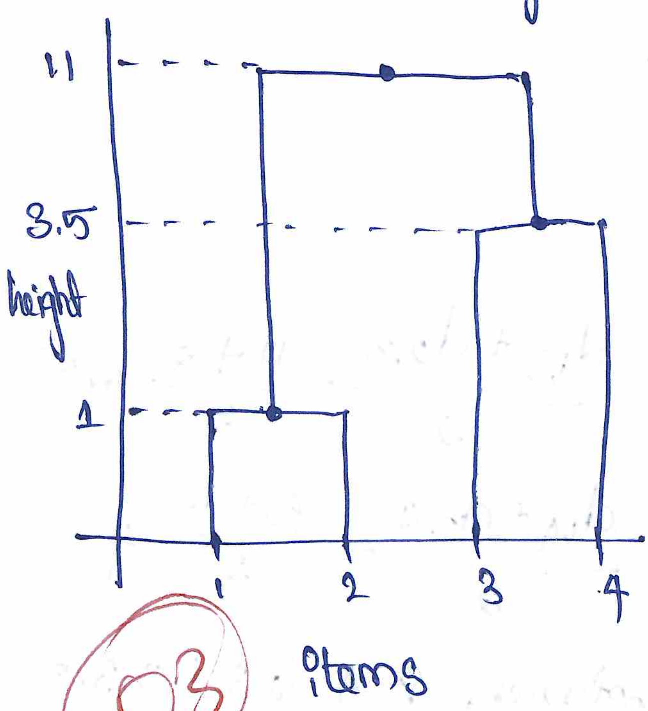
Now minimum distance is between {1,2} & {3,4} i.e 5.25

cluster 3 = {1,2,3,4} height = 5.25

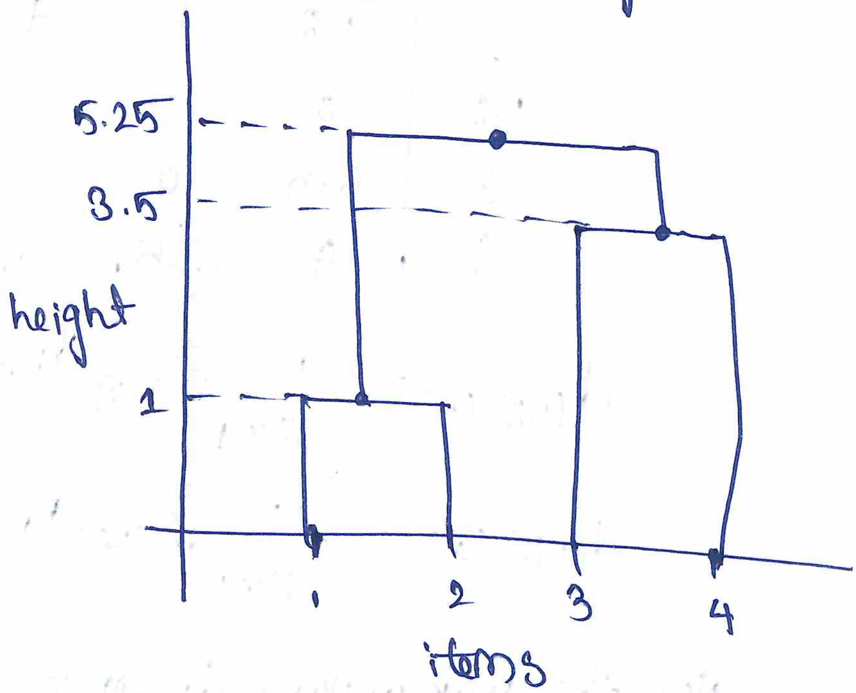
	1234
1234	0

b)

Complete linkage



Average linkage



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The heights were same when forming cluster {1,2} and {3,4} in both cases but when forming cluster {1,2,3,4} The Average linkage gives less height than complete linkage